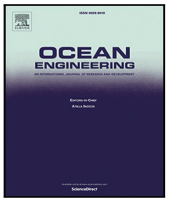
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Estimation of ship-induced sediment resuspension in intertidal waterways based on field measurements at the Hooghly River, India

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ABSTRACT

The estimation of the sediment flux due to individual ship movement is necessary to identify adverse effects on the river banks in tidally-driven waterways. The separation of each wake event due to ship propagation from the tides is cumbersome. The present study focuses on the study of the ship waves on the sediment resuspension in intertidal waterways through a field survey. The time–frequency analysis proved to be an effective tool for separating the high-frequency ship waves from the tides. The ship-generated SSC was estimated by using the backscattering intensity from the acoustic sensors after proper calibration. The individual effect of each ship wake event on the sediment resuspension was estimated by filtering the ship-generated SSC from the total SSC by using a moving average filter. The overall contribution of the ship waves during the survey tenure was 24.66% with respect to the ambient SSC. The approximate amount of sediment flux by ship waves in a calendar year could be $1.46 * 10^6 g/m^2$. Finally, a new empirical equation to predict the sediment flux from the depth Froude number is proposed, which could be used in long-term analysis of SSC after proper calibration and validation.

1. Introduction

Streambank erosion is a natural process and necessary to the health of fish-producing systems providing spawning gravels and stream morphology necessary to maintain all life stages. Many factors, such as climate, geology, and land use, control the rate at which a stream will erode its banks. When streambank erosion is accelerated above standard rates, controversy often develops about the cause of the increased erosion and how best to mitigate it. The cardinal causes of streambank erosion include the depth, river currents, movement of marine vehicles, approach angle, and sediment content of the river; the type and density of vegetation; the height and slope of the banks; the soil type; and the size of particles making up the potentially eroded material. Excessive streambank erosion could result in the failure of jetty and other berthing structures causing considerable loss of land, property and wealth of mankind. Moreover, Increased turbidity due to sediment resuspension has been identified as an important issue in river ecosystem management. When compared to clear water conditions, moderate turbidity has been found to be advantageous for predators because it improves the contrast between the water and prey. On the other hand, high levels of turbidity can reduce foraging effectiveness (Kucera-Hirzinger et al., 2009; Utne-Palm, 2004). Moreover, the increased turbidity reduces the light permeability, retarding the growth of marine phytoplanktons and increasing the biological oxygen demand in the river (Schallenberg and Burns, 2004). Therefore, it is important to study the ship-generated waves and induced sediment resuspension.

The inland waterways are one of the natural ecosystems and act as a cheaper mode of transport of bulk materials. The increasing focus on transporting bulk materials via commercial vehicles in inland waterways has imposed a major threat to the biodiversity of the inland waterways. The vessel movement in intertidal areas disturbs the natural flow field and creates complex hydrodynamics along with tidal forcing. Deep draft vessels tend to generate higher drawdown and wave height than small draft vessels (Parchure et al., 2001). Generally, the ship waves affect the river channel by turbulence and bank erosion (Bauer et al., 2002; Kevin and Parnell, 2001; Soomere, 2005). The turbulence tends to re-suspend the sediments from the bed (Soomere, 2005), posing an immediate threat to the inland waterway stability. The energy dissipation due to wave breaking affects the riverbanks and water quality (Asplund, 2000).

The ship waves tend to travel as a series of wake trains comprising of primary and secondary waves (Bertram, 2012). The primary waves are mainly generated due to water displacement because of the hull. In contrast, the secondary waves are associated with the disturbance

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from the bow and stern. The primary waves have a larger time period and reach the bank initially, causing erosion. The secondary waves consist of divergent and transverse waves, where transverse waves have relatively more minor wave height, implying that only divergent waves are responsible for sediment resuspension near the bank (Larson et al., 2017). The critical parameters influencing the ship waves are vessel speed, water depth and vessel length (Sorensen and Weggel, 1984). While each passing vessel generates a wake train with unique characteristics, the maximum wave height can be predicted from the vessel speed and depth based Froude number (Gates and Herbich, 1977; Bhowmik et al., 1981; Sorensen and Weggel, 1984; Chakraborty et al., 2022), expressed as:

$$Fr_d = \frac{V_{ship}}{\sqrt{gd}} \tag{1}$$

where V_{ship} is the speed of the vessel, Fr_d is the depth Froude number, d is the water depth, and g is the acceleration due to gravity. Due to limited water depth, vessels navigate in the sub-critical range in inland waterways. As a result, both divergent and transverse waves significantly affect sediment resuspension. It is worth mentioning that the sediment resuspension depends on the distance of the bank from the sailing path as waves decay exponentially away from the sailing line (Kofoed-Hansen, 1999). Therefore, primary and secondary wave height can significantly demonstrate the erosive potential of the wake train.

As the ship waves propagate towards the riverbanks, they result in high bed shear stress as river currents increase due to shoaling and local bathymetry orientations, causing resuspension of fine sediment particles (Gelinas et al., 2013; Houser, 2010; Parnell et al., 2008; Rapaglia et al., 2011; Schoellhamer, 1996; Soomere, 2006; Soomere et al., 2009). Contrary to relatively slowly developing wind-induced waves, flow velocities under ship waves can increase by order of magnitude within seconds (Rapaglia et al., 2015), resulting in a less gradual, almost instantaneous increase in suspended sediment concentrations (SSC). Therefore, ship waves are generally more harmful than wind waves in intertidal areas because of shorter fetch and limited water depth (Mcconchie and Toleman, 2003; Maynord, 1996). The analysis of the sediment resuspension due to individual ship wave effect has been conducted by various researchers (Osborne and Boak, 1999; Kevin and Parnell, 2001). However, quantification of the sediment resuspension due to various commercial ships (deep draft) in the intertidal environment is not reported in the literature. Studies on sediment suspension due to tides and wakes, reported by Safak et al. (2021) concluded that there is a 12% increase in sediment transport due to ship waves by analyzing the river conditions during boat activity and inactivity period. It is very difficult to separate the ship wave data from the tides. Moreover, the individual effect of each vessel on the sediment resuspension in an intertidal environment is difficult to analyze. They have not been discussed in detail in the literature to the best of the author's knowledge. As a result, there is a strong research gap in the analysis of sediment resuspension due to ship waves in intertidal zones. Therefore, a field study was conducted in the Falta region of the Hooghly river, India, to understand sediment resuspension due to commercial vehicles.

The Hooghly river is the national waterway 1 of India and home to two major ports, Syama Prasad Mookerjee Port (SMP), Kolkata and Haldia Dock Complex (HDC). It is the arm of the river Ganga, extending from Nabadwip to Sagar islands in West Bengal, India. The total length of the river is 260 km. It is a tide-dominated river with tidal forcing being semi-diurnal in nature. The tide ranges from 1.5 m to 6 m in the river. The river experiences varying bathymetry throughout the channel with the water depth ranging between 3 m and 15 m. The river received a freshwater discharge of 2590 m^3/s from the upstream (Chakraborty et al., 2022). The river experienced a traffic volume of 3518 commercial vehicles in the calendar year 2019–2020, which makes it the perfect location to study the effect of ship waves along with tides. There are certain stretches in the river where the ship navigates close to the river bank causing severe bank erosion (see Fig. 2(a)).

The primary focus of the present research is to study the effect of ship-induced sediment transport in the intertidal environment of the Hooghly river. Chakraborty et al. (2022) reported that spectrogram analysis is an effective tool in understanding the overall characteristics of the ship-generated waves in the Hooghly river. A similar analysis will be deployed in the present research to understand the frequency content within the collected time-series data. Based on the frequency difference, the ship waves will be extracted from the tides by using a moving average filter. The SSC associated with the individual ship movement will be estimated based on the sonar principle and data collected from water samples. The main assumption of the present research is that the overall river condition is considered to be constant during the ship navigation in the study domain. The paper starts with a brief summary of the study area. The details of the field experiment campaign and analysis procedure are discussed next. The results related to the comprehensive data collected, extraction of the ship waves, and estimation of SSC due to individual ship movement are reported next. The following section presents discussions on the impact of ship waves and tidal forcing on the suspended sediment concentration. Finally, a brief conclusion about the present study is presented.

2. Method

2.1. Study area

The details of the field measurements are reported in Chakraborty et al. (2022). However, a brief review of the study area is provided for the completeness of the paper. The field measurements were performed at the left bank of the Hooghly river in the Falta region, West Bengal, India, between 25th August and 2nd September 2021. An overview of the Hooghly estuary is shown in Fig. 1. The Hooghly river is a tide-dominated river with its effect extended to 175 km upstream of the river up to the Nabadwip region. Nabadwip witnesses the confluence of two non-perennial rivers, Bhagirathi and Jalangi, which discharge fresh water into the Hooghly river. The normal freshwater discharge from June to October is about 3000 m^3/s and from November to May is around 1000 m^3/s (Sadhuram et al., 2005). The mean discharge during the survey tenure was 2590 m^3/s. The river forms a tributary, the Rupnarayan river, 5 km downstream from our measurement station. The red mark shows the study location with the SMP port upstream and HDC downstream. The ships navigating towards the SMP port from the ocean would travel through our study location. The site station was set up in the Falta jetty (22° 15′17.57″N, 88° 5′16.19″E,), displayed in Fig. 2(b). The jetty protruded 76.5 m from the bank into the river. The river width near the Falta jetty is approximately 1363 m. The eroded bank profile (captured during low tide) is mainly due to the influence of the ship waves navigating close to the river bank and the tides. The vessels navigating this zone are around 105–158.98 m in length, 18.8–24.5 m in width, and 4.2–7.1 m in the draft. Apart from these deep draft vessels, various barges and small vessels navigate the region throughout the day. The vessels usually navigate during the high tide due to the non-availability of adequate water depth in the river channel. The ships navigate very close to the left bank and even closer to the Falta jetty.

The tidal characteristics in the study domain are semi-diurnal, ranging from 1.1 m to 6.6 m during the spring tide. The river bathymetry profile at the marked cross-section in Fig. 1 is reported in Fig. 3. The water depth at the measurement location (denoted by A in Fig. 3) is 1.63 m with respect to the chart datum, which is 2.82 m below the mean sea level. The mean high and low tide during the survey tenure was 5.85 m and 1.63 m, respectively, denoted by the magenta dashed line in the figure. There is a sand bar formation (see Fig. 1, 3) at the right bank, which gets completely submerged during the high tide. The approximate bank profile is represented by the red dashed line, which however could not be measured due to the lack of space availability



Fig. 1. The overview of the Hooghly river with the Bay of Bengal (downstream) and Syama Prasad Mookerjee Port (upstream). The aerial view shows that the Hooghly river has a tributary, the Rupnarayan river, 5 km downstream of the measurement location. The enlarged view of the study site is shown in the same Figure. The field measurements were observed in the Falta jetty (22° 15′17.57″N, 88° 5′16.19″E,), where the river width is approximately 1363 m.



(a) (b)

Fig. 2. (a) Typical scenery of bank erosion in the left bank of the river Hooghly in Falta region. A vertical cut of approximately 1,5 m was noticed due to tidal fluctuations and ship navigation. (b) Side view of the Falta jetty with the station setup. The bank is severely eroded with time due to the influence of tide and ship waves. Both images are taken during the low tide.

of the launch vessel. The ships usually navigate during the daytime at high tide due to the lack of adequate depth in the study location. The average river currents in the study domain range from 0.8 m/s to 1.5 m/s, with the maximum value at 2.5 m/s during the spring tide. Most of the sediments in the region are non-cohesive sand with a bulk density of 2650 kg/m^3 (Arora and Bhaskaran, 2013). The mean sediment diameter in the Falta region is 0.12 mm, obtained from the SMP port database. The Hooghly river is highly dynamic in terms of sediment load-carrying capacity into the sea and carries about $616 * 10^6$ t of suspended sediments into the Bay of Bengal (Qasim et al., 1988). According to Lisitzin (1972), the Hooghly estuary is the second largest river in the world in terms of suspended sediment contribution to the deep sea.

2.2. Data collection

The instruments used to measure ship waves, tide, river currents, and backscatter intensity were mounted adjacent to each other on the Falta jetty. Adequate spacing was maintained in the mounting setup of each instrument to avoid the interference of one instrument with the other. The continuous profile of the water surface elevation was captured using an ultrasonic wave-tide recorder (LOG aLevel®, General Acoustics) at a sampling frequency of 5 Hz. The recorder protruded 1 m outside from the jetty top to minimize the interference from the jetty (see Fig. 4(a)). The instrument is equipped with a reference sensor for self-calibration. A capacitance wave gauge of 1 m in length (usually used in the laboratory) was mounted beside the ultrasonic recorder to cross-verify the results from the recorder. The wave gauge was calibrated every day before the passage of ship traffic. The comparison of the water surface elevation from the ultrasonic recorder and wave gauge is reported in Chakraborty et al. (2022).

A horizontal-looking Acoustic Doppler Current Profiler (HADCP) (300 kHz, Channel Master, Teledyne RD Instruments) was installed beside the wave gauge, as displayed in Fig. 4(b). The HADCP was installed 4.5 m above the river bed. The mean high water level at the

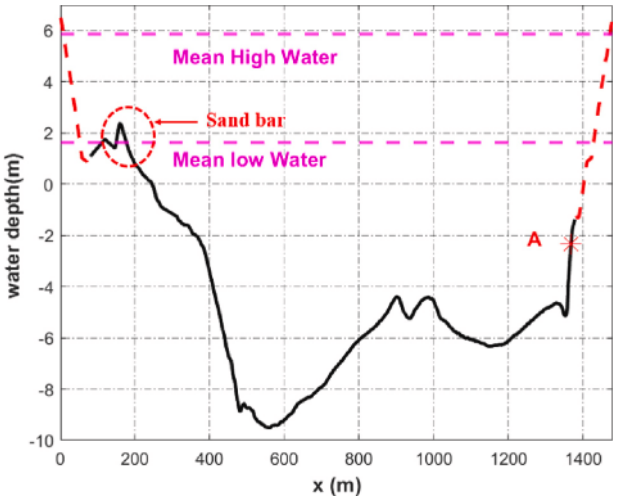


Fig. 3. Cross-sectional profile of the river bed with respect to the charted datum, which is 2.82 m below the mean sea level (MSL). Point A denotes the measurement location of the ship waves and tidal elevation. The mean low and high water during the study period is shown by the magenta dashed line. The red dashed line represents the approximate bank profile (could not be captured during bathymetry due to limited space availability of the launch vessel), which is submerged during high tide and emerges during low tide. The approximate bank profile is visually inferred from Fig. 2(b).

measurement point was 6 m. Therefore, considering the high water level, the ADCP was placed at the mid-depth of the channel during ship propagation. The deployment of the instrument at the mid-depth is ideal as the sensors could neglect the river bed's interference and capture the ship waves' influence without the beam projecting outside the water column. The ADCP logged at a sampling frequency of 2 Hz and gave an output every 5 s by averaging the collected data. The data was averaged to minimize the fluctuations and errors due to turbulence. The data was collected during the daytime until the ADCP sensor came out of the water. The velocity data obtained from HADCP were cross-verified manually by calculating the time required by a floating object to pass between two reference points. The velocity data obtained from the HADCP gave a good overview of the river velocity variations during the spring and neap periods. The instrument revealed daily velocity variation in the river from flood to the ebb phase and captured the influence of ship waves. Since the ships navigate during high tide and daytime alone, the ADCP data was not collected at night. The backscattering intensity measurements obtained from the HADCP could provide sufficient information about SSC by calibrating the echo intensity levels with in situ collected water samples.

The bathymetry profile along the river cross-section was measured using Bathy 500 MF echo sounder and DGPS installed in a survey launch vessel MV Sarthak. The data was collected in a pre-defined line profile of 200 m spacing for a stretch of 2 km upstream and downstream of the measurement location. The data was collected during the high tide to cover the entire river width. However, bathymetry data near the jetty was impossible to collect due to the lack of space for vessel navigation and the non-availability of water depth. Thus, the bathymetry data near the jetty was evaluated by linear interpolation from nearby data points. The list of the ships navigating through the study location was obtained from the SMP port. Moreover, the vessels' incoming and outgoing timings were well documented for easy identification and extraction from raw data. The important ship parameters such as ship length, beam, draft and speed were collected from MarineTraffic (2021). The shipping speed was cross-verified from the (MarineTraffic, 2021) data by calculating the time required to pass between two reference points marked in the field. The updated coordinates of the ship navigation at the study site (obtained from MarineTraffic, 2021) gave the ship's sailing path distance from the Falta jetty.

2.3. Data processing

While the generation of the ship waves in the deep waters is well understood and documented (Havelock, 1908; Thomson, 1887; Sorensen and Weggel, 1984), the slight variation in the sailing regime and local bathymetry generates complex wave pattern in shallow intertidal waterways (Stumbo et al., 1999; Chakraborty et al., 2022). As these waves are comprised of linear and non-linear components (Didenkulova et al., 2013), appropriate methods should be deployed to separate these waves from the tidal forcing. Ships traveling at high velocity generate non-stationary components. Under these complex conditions, the traditional stationary process (ergodic) and statistical methods fail to characterize the ship waves and extraction of the ship waves from the tidal fluctuations is complex. Therefore, using time–frequency analysis could be a valuable tool for identifying ship waves. Sheremet et al. (2012) reported that the ship waves could be identified from strong wind wave background using time–frequency analysis. Owing to the finite spatial extensions (Torsvik et al., 2009), transient nature and short timescale, the ship waves appear as chirp signals (frequency increasing with time) in the time domain. Therefore, by applying spectrogram analysis to the measured time series, the ship wakes were identified by monotonically increasing peak frequency, where the energy content was highest. The list of the ships navigating through the study site was available prior to the movement. The real field documentation of the ship's in and out time at the Falta jetty was carried out to connect the wake measurements with the individual ships. The raw data was processed using the computer software LOG aLevel® Player.

After identifying individual ship waves in the measured data, the velocity and acoustic backscattered data at the same time interval of ship navigation were extracted from the HADCP (by using Channelmaster Utility software). The ADCP data were pre-processed by removing the noisy data with a low signal-to-noise ratio (less than 40 dB), and the missing data was calculated by linear interpolation. The backscattering profile obtained from the HADCP was calibrated using the in-situ water samples collected by Niskin Sampler (2.5 L) near the jetty, from the high tide to the low tide time at an interval of 1 h. The suspended sediment concentration for all the wake events was therefore estimated, by utilizing the empirical equation obtained from calibration. Similar works are reported in Venditti et al. (2016). The method to separate the ship-induced SSC from the tidal influence and upstream discharge is discussed in the later section. The salinity measured by the CTD sensor was negligible in the study domain. The measured bathymetry profile was processed in the Hy-pack software, and the corresponding tide data was neglected to obtain the river bed profile (see Fig. 3).

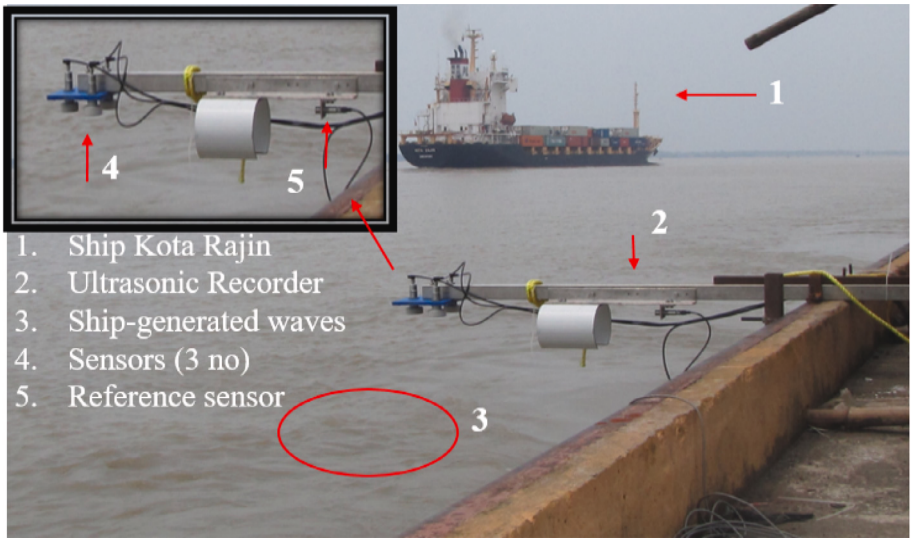
2.4. Sediment fluxes

The sediments resuspended by the individual ship movement are carried downstream with the river currents and get deposited at specific places. This eventually increases the overall dredging cost of the river. Therefore, the estimation of the sediment flux is an important parameter for understanding the overall sediments contributed by the ship movement. The sediment rate could be estimated by using the following equation (Ravens and Thomas, 2008):

$$q_s = u * C_s \tag{2}$$

where q_s is the sediment rate, u is the horizontal velocity measured by the ADCP and C_s is the suspended sediment concentration analyzed using the backscattering intensity of the ADCP. The calculation of the analysis of the SSC using acoustic data from ADCP is reported in Appendix. The integral of the sediment rate over the time required by a particular ship to traverse the study domain gives the direct estimation of the sediment flux by individual ship movement. The sediment flux is given by:

$$SSC_{flux} = \int_0^t q_s dt \tag{3}$$



(a)



(b)

Fig. 4. (a) Set up of the ultrasonic recorder. The water surface fluctuation due to the ship Kota Rajin is being captured by the instrument. The sampling frequency is 5 Hz, (b) Setup of the Horizontal looking ADCP from the jetty top. The ADCP is mounted beside the wave gauge with adequate clearance to neglect interference between each other. The image is captured during low tide time.

where t is the time traversed by each ship through the study domain and SSC_{flux} is the sediment flux. The sediment flux is calculated by using numerical integration (trapezoidal rule) in the MATLAB tool. The integral of the sediment flux gave a good overview of the sediment estimation with respect to the temporal scale. It should be noted that the cross-profile velocity is not considered in the present analysis as it was found to be minimal with respect to the horizontal velocity.

3. Results

3.1. Overview of collected data

The field study was conducted during the spring and neap periods to understand the ship's hydrodynamics under different tidal conditions. In the total tenure of the field survey, 25th August to 27th August was the spring period and 28th August to 2nd September was the neap period. The readings on 1st September was not captured due to a power failure at the site station. In order to manifest a glimpse of the site conditions during the spring and neap periods, the overview of the collected data is reported in Fig. 5. Semi-diurnal tidal conditions are prevalent in the site condition. The highest water level fluctuations were noticed on 26th August in the spring period with a low water level at 1.1 m and a high water level at 5.95 m (see Fig. 5(a)). Similarly, the water level fluctuations were much less during the neap period, with the lowest variation experienced on 31st August, having a low water level of 2.63 m and a high water level of 4.45 m. The tide data shown in the figure is with respect to the Chart Datum (CD), which is 2.82 m below the mean sea level (MSL). The water level fluctuations reported in Fig. 5 are consistent with the nearby tidal gauge of the SMP port.

The current velocity was collected during the daytime as the relevant water level was not available for HADCP during the nighttime. Since the ADCP was mounted on the left bank, the direction of flood currents is positive, and the direction of ebb currents is negative, as per the instrument coordinates. Fig. 5 reveals that the current flow is higher

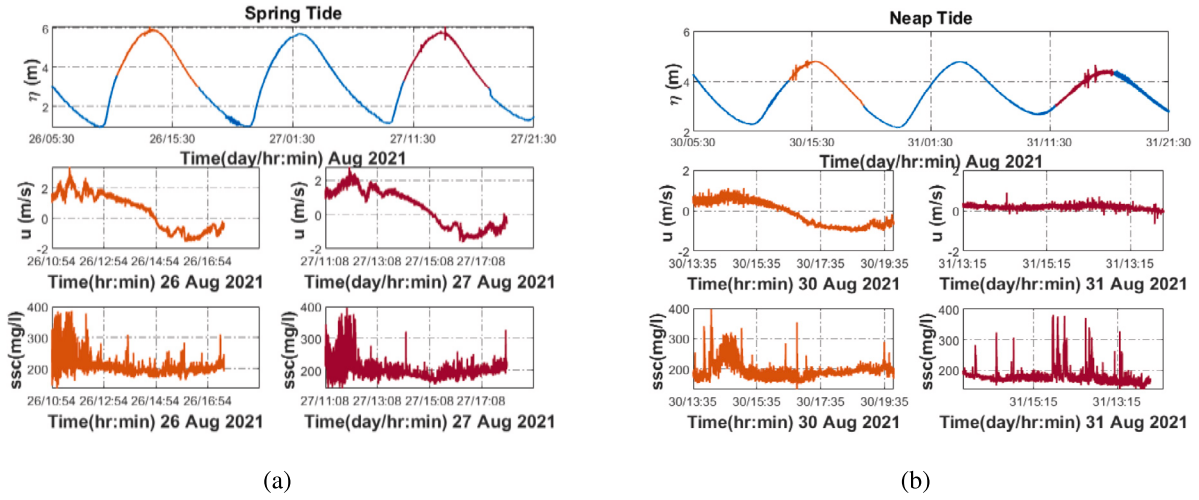


Fig. 5. The general characteristics of the site condition during (a) 26th and 27th August 2021 in the spring period and (b) 30th and 31st August 2021 in the neap period. η is the overall water surface variation in the study area corresponding to tidal fluctuations, u is the current speed and SSC is the suspended sediment concentration obtained at 4.5 m above the river bed. The velocity profile was measured during the daytime until the ADCP sensor came out of the water. The water level fluctuations are with respect to the Chart Datum level.

during the spring period compared to the neap period due to large tidal fluctuations. The average river currents during the spring period is 1.5 m/s with the maximum value of 3.0 m/s observed on 26th August 2021 (see Fig. 5(a)). The river conditions are much calmer during the neap period compared to the spring period, with average river currents of 0.5 m/s and the highest observed river current was 1.1 m/s (see Fig. 5(b)). The sediment concentration was much higher during the spring than in the neap period. The sediment concentration during the flood phase in the spring period was approximately 396.55 mg/l, whereas the sediment concentration during the flood phase in the neap period was around 230 mg/l.

3.2. Identification of ship waves

A spectrogram study was used to separate the ship waves from the tides. The analysis was carried out with the spectrogram function in Matlab with a Hamming window of 180 s and a window overlap of 170 s. The results were reported in a time–frequency map with frequency on the y-axis and time on the x-axis with color intensity of $\log_{10}(PSD)$, where PSD is the power spectral density. The spectrogram analysis of the raw data, including both tide and ship waves on 30th August 2021, is reported in Fig. 6. The entire spectrogram analysis of the raw data during the daytime at the start of the flood phase to the onset of the ebb phase is shown in Fig. 6. The time-series of the water surface elevation shown in Fig. 6 confirms the presence of low period water surface fluctuations along with high period tides. The database containing the list of ships passing through the study domain confirmed the presence of ships during this time. Moreover, The monotonically increasing frequency appears as the chirp signal in the spectrogram plot (see Fig. 6). These transient chirp signals cause an instantaneous increase in the energy within the spectrogram plot and dissipate very quickly. It is known from the literature (Didenkulova et al., 2013) that ship waves appear as transient chirp signals in the spectrogram plot. Moreover, the ship-generated waves attenuate much faster as they dissipate along the river cross-section. Therefore, by using the time–frequency analysis, two different frequency zone were identified in the spectrogram plot. The high-frequency signals were due to the presence of ship waves whereas the high-energy low-frequency signal was generated due to tides. Based on the frequency difference, a moving average filter has been applied to the raw data set to separate the ship-generated waves from the tides.

The two peaks on the time series of the water surface elevation and corresponding chirp signals between 13:51 and 14:13 are identified

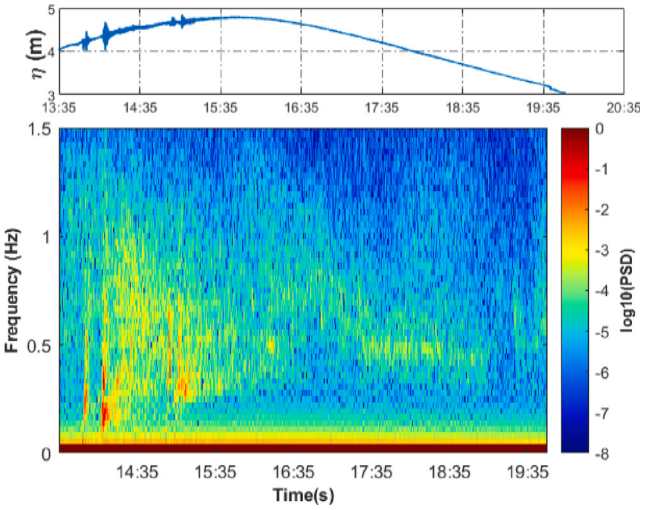


Fig. 6. Time series of the water level fluctuations along with the spectrogram plot on 30th August 2021.

as ship Sol Progress and Indian Coast Guard ship (ICGS). Similarly, another set of two peaks between 14:57 and 15:15 are identified as ship Ratha Bhum and Guang Ping. An entire 8-minute time series during the individual ship passage was selected and detrend to neglect the effect of tides. The comparative spectrogram plot during the passage of ship Guang Ping and the no-ship period is illustrated in Fig. 7. The maximum intensity of the spectrogram in Fig. 7(a) can be found at the fold of the two branches, which corresponds to divergent and transverse waves (Torsvik et al., 2015). Similarly, there is a substantial absence of high energy in Fig. 7(b). However, the small intensities in Fig. 7(b) are due to background noise and wind waves, which can be neglected by performing a low pass filter on the raw data. Wind waves are minimal in this region due to short fetches in the study domain. Therefore, spectrogram analysis was an effective method to identify individual ship waves from the raw data set (comprised of tidal and ship waves). Similarly, this method was deployed to identify the individual ship waves for the entire data set obtained during the field study. A total of 30 deep-draft commercial vessels were identified in the entire tenure of the field survey. However, as ships navigate only during high tide, ships often cross each other in this region, and waves get mixed. Therefore,

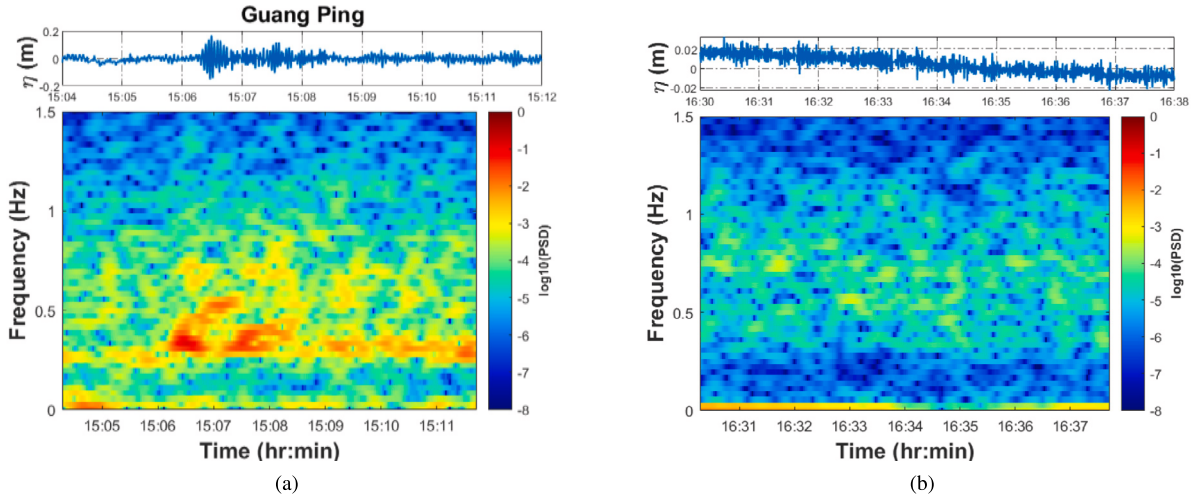


Fig. 7. Spectrogram plot of 8-minute de-tided data of (a) ship Guang Ping (b) calm period (no ship activity). The high intensity in plot (a) is due to ship Guang Ping, while in a strong absence of high intensity in plot (b).

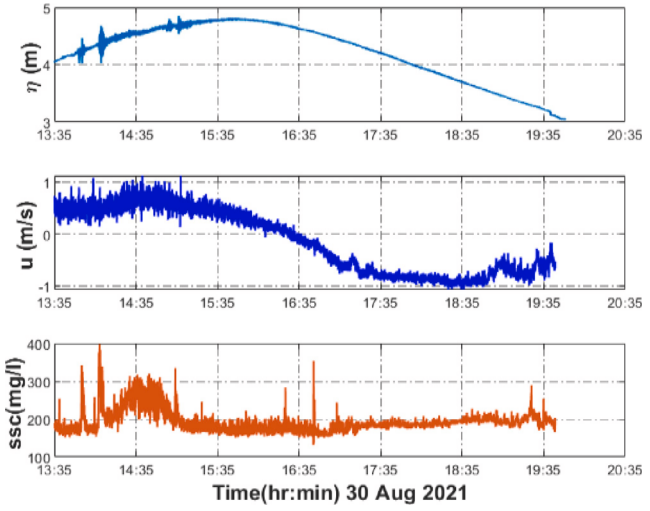


Fig. 8. Plot of the water surface elevation, velocity and suspended sediment concentration (SSC) at Falta jetty on 30th August 2021. The velocity and SSC values are measured at 4.5 m above the river bed.

individual movement of 24 ship data is considered for this study. The movement of fishing boats and barges is not considered in this study, as the individual effect of small boats on the SSC was significantly less compared to commercial vessels.

3.3. Sediment processes due to ship waves

To get an understanding of the sediment processes for a particular day, the water surface elevation, velocity, and suspended sediment concentration profile for 30th August 2021 is presented in Fig. 8. The sharp peak in the water surface elevation and suspended sediment concentration (SSC) during the ship navigation time is quite evident from Fig. 8. In contrast, a declining profile of SSC is observed during the calm water period (16:30 to 16:40), where the current speed in the river is almost 0 m/s. Thus, when the high tide has reached its peak, there is no more sediment influx from the ocean to the river. The sediment concentration during the calm water period is mainly due to the upstream discharge's contribution and is referred to as background concentration in this study.

In the previous section, we have correctly separated the ship waves from the raw data. The SSC variation during the same 8-minute time stamp (when ship waves reached the jetty) was analyzed and reported in Fig. 9. The suspended sediment concentration shown in Fig. 9(a) has a contribution from tides, ship waves and upstream discharge. It is challenging to identify the contribution due to ship waves alone. However, contrary to tidal waves, flow velocities under ship waves can increase by an order of magnitude within seconds, resulting in a less gradual, almost instantaneous increase in suspended sediment concentrations. Moreover, the SSC only increases for relatively short periods (per wave event), in order of minutes after a passing vessel, as waves attenuate much faster, while the background current of the river usually sweeps the suspended plumes downstream (Bauer et al., 2002; Smarts et al., 1985). The contribution of the ship waves could easily be filtered out by passing a moving average filter on the 8-minute time series data. After passing a moving average filter on Fig. 9(a), the plot of the suspended sediment concentration due to ship Guang Ping alone is shown in Fig. 9(b). The maximum SSC contribution of the ship Guang Ping is 130.96 mg/l. Any negative value in SSC that occurred by applying the moving average filter is discarded and assigned a 0 value, as shown in Fig. 9. This consideration is reasonable for process understanding.

The time series of SSC due to the ship Guang Ping shows the maximum SSC between 15:07 h and 15:09 h, exactly at the location of maximum intensity in Fig. 7(a). Therefore, it could be said that we have correctly analyzed the SSC contribution due to the ship Guang Ping. In order to verify our understanding, the SSC due to ship Guang Ping is now compared to the SSC during the calm water period when there is no ship activity. There is no significant instantaneous rise in the SSC during the calm water period (see Fig. 9(c)) similar to our findings in the spectrogram study (see Fig. 7(b)). Therefore, we could say that the current methodology correctly identifies the ship movement in our study site and analyses the individual SSC contribution due to ship movement. The present analysis also helps to identify the overall background SSC during the calm water period when there is no ship activity. Therefore, the same methodology is being applied to the entire data set. The individual contribution of each ship, along with the tidal influence, is reported in Table 1.

It should be noted that the ADCP was designed to measure data up to a distance of 300 m. The navigation of the deep-draft vessels in the study location is pre-defined and authors have no control over it. It was noticed that the ships navigated between 150 m and 250 m from the Falta jetty. The navigation of the vessels in the study domain was pre-defined and authors have no command of the navigation path followed by each ship. This was the primary reason behind choosing the particular ADCP for the present research. However, it should be noted

Table 1
Daywise list of the maximum SSC during flood phase, ebb phase, calm water phase and individual ship movement. The up-arrow denotes the movement of the ship toward SMP Port while the down-arrow denotes the movement toward the ocean.

Date	Flood phase SSC (mg/l)	Ebb phase SSC (mg/l)	Calm phase SSC (mg/l)	Ship name	Ship induced SSC (mg/l)	Tide
25/08/2021	384.5	316.08	178.4	SSL Chennai ↑ Imke Schepers↓ SSL Sabarimalai↓	9.49 8.64 6.3	Spring
26/08/2021	385.44	311.0	185.3	Xpress Meghna↓ Ark Progress↑	17.16 22.33	Spring
27/08/2021	396.55	247.17	181.06	IZUMO↓ UNISKY↑ Xing Ping↓	13.82 26.44 19.93	Spring
28/08/2021	364.46	304.5	170.05	Ark Progress↓ Kota Rajin↓ Kang Ping↑ SSL Chennai↓	53.2 20.97 91.18 24.36	Neap
29/08/2021	329.63	272.213	157.45	Supa Bhum↓ Reinhard Schepers↑	23.46 80.8	Neap
30/08/2021	261.08	228.82	154.45	Sol Progress↓ Ratha Bhum↑ ICGS↑ Guang Ping↑	117.85 30.26 138.82 130.96	Neap
31/08/2021	196.96	171.11	136.35	Kang Ping↓ UNISKY↓ ICGS↑	136.87 159.63 166.65	Neap
02/09/2021	163.65	218.69	156.57	MTT Senari↓ Guang Ping↓ Asiatic Cloud↑	132.5 17.66 76.89	Neap

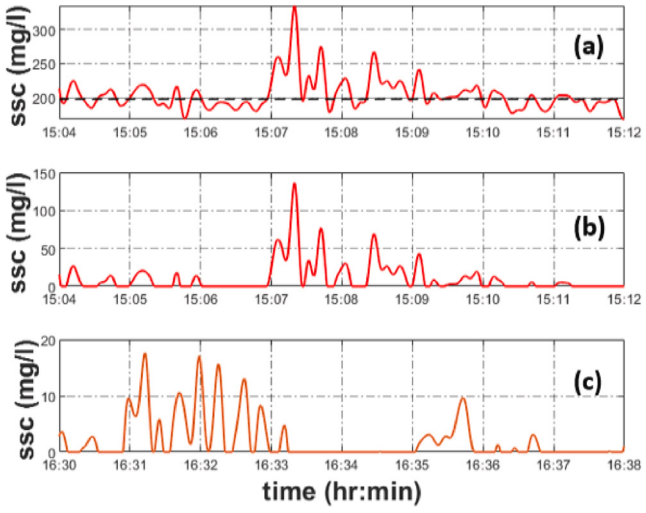


Fig. 9. Plot of the 8-minute time series of (a) the raw data of SSC during the navigation of ship Guang Ping at our study site. The black dashed line is the moving average filter applied to the raw data. (b) the individual contribution of SSC due to the ship Guang Ping. (c) the SSC during calm water period when there is no ship activity.

that the end distance of data measurement by the ADCP is dependent on the signal-to-noise ratio. Therefore, at high sediment concentrations, the ADCP could not collect data up to 300 m and it was reduced to 150 m. The study domain experienced a huge sediment influx during the onset of tides. Therefore, during the starting of flood or ebb tide, the ADCP measured data up to 150 m in the spatial scale. Since our analysis is mainly at the measurement point A, near the Falta jetty, this limitation of the ADCP did not affect our analysis. Moreover, the ADCP could measure data up to 300 m during the time of ship passage, as tidal influence had been stabilized by that time and there was no sediment influx from the ocean. Another limitation of the ADCP was that it could provide data output every 5 s after averaging data collected at a sampling rate of 2 Hz. This was mainly done to minimize the noise level in the ADCP. As the data was collected every 5 s, which could be ideal for tide measurements, however, the data collected is slightly less to represent the high-frequency ship waves. Therefore, the velocity data and backscattering intensity obtained for the ship-generated waves may be slightly underpredicted in the present study.

The analyzed SSC data obtained for each ship wake event is site-specific. The SSC obtained by using the calibration process has been explained in detail in Chakraborty et al. (2023b). However, due to the standalone nature of the present paper, the details are briefly reported in Appendix. The calibration of the backscattering intensity was performed based on field-measured SSC. Therefore, the present method is dependent on site-specific conditions such as sediment size and water depth. The d_{50} value for the present case study is 1 mm for the Falta jetty (Chakraborty et al., 2023a). So further investigation of the SSC with the cardinal ship parameters and river conditions is required to understand the impact of the ship-generated waves on sediment resuspension in restricted waterways.

3.4. Estimation of sediment fluxes

The estimation of the sediment fluxes is important as it helps to understand the erosive potential of the wave train. The estimation of the sediment flux pertaining to individual vessel movement is not reported in the literature, true to our knowledge. Therefore, the overall estimation of the sediment flux per wave event helps to understand the contribution of a particular ship to the bank erosion in the Hooghly river. The analysis of the sediment flux was carried out by the methodology discussed in Section 2.4. The estimated sediment flux by the individual ship movement is listed in Table 2.

It is clear from Table 2 that the sediment flux generated by the ships during the spring tide is less compared to the ships during the neap tide. The reason behind such contrasting features is explained in detail in the next section. A correlation study between the cardinal ship parameters and sediment flux was performed to understand the erosive potential of individual wake trains. The non-linearity associated with the ship passage through the study domain increases with the increase in the depth Froude number and ship speed (V_{ship}) (Chakraborty et al., 2022). The increase in the non-linearity would increase the overall ship-generated

Table 2
Summary of the important ship parameters and maximum sediment flux induced by ship-generated waves.

Ship name	Ship speed V_{ship} (m/s)	Mean time period T_{mean} (s)	Depth froude number Fr_d	Sediment flux SSC_{flux} (g/m^2)
SSL Chennai	6.17	2.48	0.702	50.94
Imke Schepers	6.17	2.52	0.696	190.93
SSL Sabarimalai	5.92	2.31	0.669	67.48
Xpress Meghna	5.65	2.50	0.654	148.13
Ark Progress	5.65	1.92	0.647	113.82
IZUMO	5.14	3.34	0.632	502.61
UNISKY	5.65	1.97	0.664	151.92
Xing Ping	6.68	2.63	0.767	119.57
Ark Progress	5.65	2.33	0.701	270.69
Kota Rajin	5.40	2.60	0.66	122.17
Kang Ping	6.68	2.47	0.805	380.43
SSL Chennai	6.17	2.07	0.73	143.57
Supa Bhum	5.40	1.71	0.66	199.56
Reinhard Schepers	6.74	1.82	0.82	151.82
Sol Progress	5.91	2.26	0.77	928.43
Ratha Bhum	6.53	1.84	0.724	681.92
ICGS	7.72	3.98	0.99	1342.2
Guang Ping	5.92	2.44	0.792	420.18
Kang Ping	6.17	1.91	0.82	1053.5
UNISKY	5.97	1.84	0.789	555.77
ICGS	8.23	3.65	1.08	1136.8
MTT Senari	5.65	2.16	0.718	577.23
Guang Ping	6.17	2.78	0.786	126.65
Asiatic Cloud	6.17	2.13	0.788	551.00

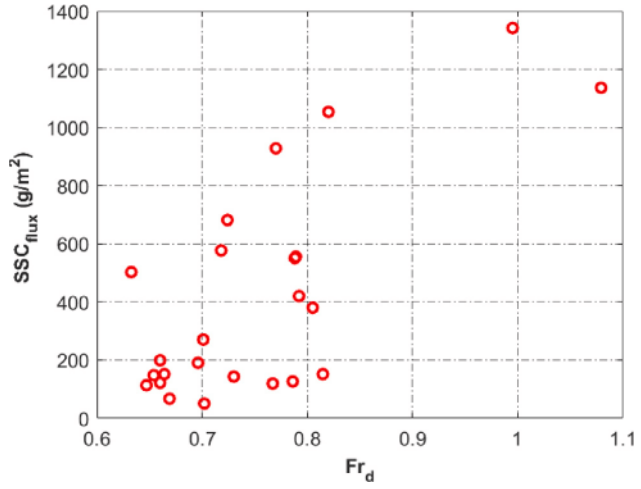


Fig. 10. Results of the comparative analysis of the sediment flux with increasing depth Froude number.

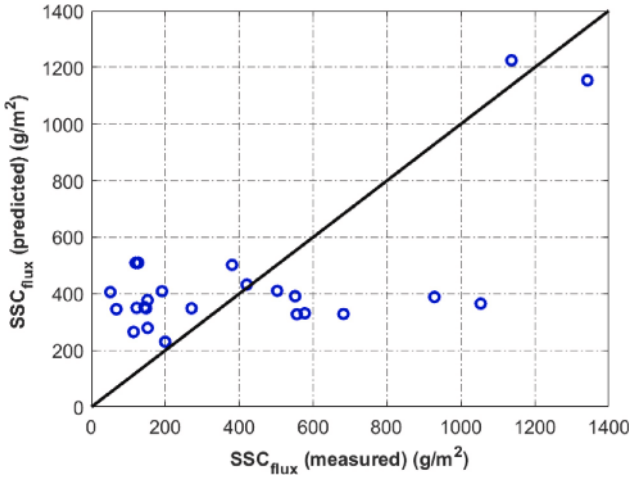


Fig. 11. Results of the comparative analysis of the predicted sediment flux with the field analyzed sediment flux.

hydrodynamics with the channel. This will eventually increase the sediment concentration in the river. The present understanding was justified as the correlation study revealed that the sediment flux generated by the ship is highly correlated with Fr_d with a correlation coefficient of 0.73 between them. The correlation analysis also revealed that the mean time period (T_{mean}) is influential in the generation of sediment flux. The correlation coefficient between the sediment flux and T_{mean} is 0.43.

Fig. 10 reveals that the sediment flux increases with the increasing depth Froude number. Therefore, based on the current dataset, new empirical relation is formulated to predict the sediment fluxes based on cardinal ship parameters by using regression analysis. The empirical equation is given as follows:

$$SSC_{flux}^{pred} = 1.43 \times 10^{-5} \rho_s V_{ship} T_{mean} Fr_d \quad (4)$$

where SSC_{flux}^{pred} is the predicted sediment flux and ρ_s is the sediment density in g/m^3. The sediment density for the present case study is 2650000 g/m^3. Statistical analysis between the predicted and measured sediment flux revealed that they are highly correlated with each other with a correlation coefficient of 0.65 and coefficient of determination of 0.5. The comparative analysis of the measured and predicted data is shown in Fig. 11. Eq. (4) slightly overpredicts the data as the non-linearity associated with each vessel movement could not be implemented efficiently in the simplistic empirical equation. Moreover, the present model holds good only for deep-draft vessels. Detailed investigation and field measurements of different ships at varying bathymetry of various inland waterways could enhance the efficiency of the empirical model. This is kept as a future scope of work. However, the proposed equation is useful for a preliminary understanding of the erosive potential of each wake event due to ship movement.

The rough estimation of the sediment flux contribution due to ship waves is reported here. As mentioned earlier, the river experienced an approximate traffic of 3518 vessels for the calendar year 2019–2020. Based on our field study, the average contribution of the sediment flux due to ship traffic is 416.2 g/m^2. Therefore, for a calendar year, the river will experience a sediment flux of $1.46 * 10^6$ g/m^2 due to the ship's movement alone without the tides in the waterway. However,

it should be noted that seasonal variability plays an important role in the variation of the sediment flux. The present field measurement was performed during the monsoon season. As a result, river discharge and sediment influx from upstream would be high compared to other seasons. It is clear from Table 1 that ship-generated SSC is less during spring tide compared to the neap tide. Moreover, in calm river conditions with less influence from tides and upstream sediment flux, the effect of ship movement on sediment resuspension would be high. Therefore, the present estimation of ship-generated sediment flux is slightly underestimated compared to other seasons. However, it should be noted that combined sediment influx due to tides, river currents and ship-generated waves is highest during this season compared to other seasons. A detailed explanation of sediment variation with respect to tides and ship-generated waves is reported in the following section.

3.5. Ship waves versus tidal forcing on sediment resuspension

Based on the above discussion, tidal forcing and ship waves are identified as the main drivers of sediment transport. A two-way flow of river currents characterizes a tidal cycle for a particular day. The flow of river currents from the ocean to the river is called flood current, while the flow of currents from the river to the ocean is ebb current. The overall channel dynamics are stable during the calm water phase (when the river velocity is almost 0 m/s). The SSC during this phase is mainly due to the contribution of the upstream discharge. This SSC can be termed the background SSC, which will be present during the flood phase, ebb phase and ship passage. The individual contribution of flood, ebb and ship-induced currents on SSC along with the background SSC is analyzed and discussed here. Coming to the contribution of tides on sediment transport, a day-wise comparative plot of the SSC during the flood and ebb phases is reported in Fig. 12(a). The entire spring period is characterized by greater SSC than the neap period due to differences in tidal fluctuations and river currents. The average amplitude of the tidal fluctuations during the spring period was 4 m as compared to the 2 m during the neap period. This observation is comparable to the measured tide data at the site location by the SMP port authorities. Higher tidal fluctuations during the spring period resulted in increased river currents (about 3 m/s), which eventually increased the sediment resuspension. The SSC due to flood and ebb currents were 111.62% and 36.6% greater in the spring period than in the neap period. Coming to the daily variation of SSC, it is noticed that the SSC induced by the flood currents is 59.9% higher than the ebb currents during the entire survey tenure. Moreover, the occurrence of tidal bores at shallow stretches during the spring period resulted in a higher variation of SSC in the flood phase than the ebb phase as seen in Fig. 12(a), compared to the daily variation of SSC in neap phase.

Now, the influence of ship waves on sediment resuspension is discussed during the contrasting condition of the spring and neap periods. The average SSC for all the ships listed in Table 1 for a particular day is evaluated and compared to the SSC values during the flood, ebb and calm water phases separately. The contribution of the ship waves on the SSC, with respect to the flood phase during the spring and neap period along with the background concentration, is displayed in Fig. 12(b). The background concentration is neglected in analyzing the influence of ship-induced SSC compared to the SSC generated due to flood currents. Due to higher tidal fluctuations and rapid river currents, tidal forcing is dominant during spring tide and responsible for most sediment resuspension. Ship-induced sediment resuspension is 7.71% of the SSC contribution due to flood currents during the spring phase. On the other hand, the ship waves contribute 115.97% of the SSC with respect to the SSC contribution of flood currents during neap tide. Therefore, ship waves are the main drivers of sediment resuspension during the neap period, and their effect is well pronounced (see Fig. 12(b).

Similarly, the comparison of SSC due to ship waves and ebb currents shows similar patterns as discussed above. The ebb currents are high during the spring phase and contribute to most of the SSC (see Fig. 12(c). The ship-induced SSC is 17.34% of the SSC contribution due to ebb currents during the spring phase, whereas contrasting characteristics are observed during the neap phase. The SSC contribution of ship waves is 157.42% with respect to the ebb currents during the neap phase. Therefore, ship-generated waves significantly control the sediment dynamics in intertidal waterways.

Thus, the contribution of the ship waves on the sediment resuspension with respect to the background suspended sediment concentration is an essential study for bank protection works and the installation of marine structures. Based on the estimation of the SSC due to ship waves, listed in Table 1, the comparative plot of the SSC with the background SSC during the calm period is reported in Fig. 12(d). The overall SSC contribution of the ship waves is 24.66% during the calm phase. Therefore, the SSC contribution of ship waves is noteworthy in intertidal waterways and should not be neglected. The day-wise percentage contribution of the ship waves with respect to the SSC contribution during the flood currents, ebb currents and the calm water phase is summarized in Fig. 13.

Fig. 13 reveals that though tidal forcing is the significant contributor of SSC in inter-tidal waterways, the contribution of ship waves is quite significant and should be considered to maintain a sustainable waterway. The tidal cycle's spring phase suppresses ship waves' contribution due to its high currents followed by high resuspension of sediments. However, the viable contribution of ship waves are well pronounced in the neap phase and is the primary driver of sediment transport in the waterway. The sediment re-suspended due to ship waves is constantly carried downstream by river currents, increasing the dredging cost in the channel as explained in the previous section.

4. Comparative study and discussion

Though a lot of studies on ship-generated waves and sediment resuspension have been performed in the past decade, a detailed study of the ship-generated waves in complex intertidal waterways has never been performed. Therefore, the primary focus of the present study was to understand the impact of ship-generated waves on sediment resuspension in an intertidal waterway. The main challenge in the present study was to separate the ship-generated waves from the tidal forcing. Safak et al. (2021) attempted to study the boat-induced sediment resuspension in an intertidal waterway. The main limitation reported in the study of Safak et al. (2021) was the filtering of the individual effect of each wake event of a boat passage from the tidal forcing. The primary reason behind the limitation was reported to be the ever-evolving nature of the river conditions. The sediment processes and bed morphology for each wake event were different. Moreover, the variation in the consolidation of the riverbed with the passage of vessels on a particular day was an issue in estimating the sediment resuspension due to individual vessel movement. The present study focuses on addressing these research gaps.

The first attempt was made to separate the ship waves with the tidal forcing. The time–frequency analysis has proved to be an effective tool in identifying the high-frequency ship waves from the tidal forcing. Fig. 6 identified three high-frequency chirp signals between 13:51 h and 15:15 h. The field measurement database was designed in a systematic way to make entries of the in-time and out-time of a particular vessel navigating through the study domain. The database entry and chirp signals confirmed the presence of ship waves in the water surface elevation data captured using the ultrasonic recorder. Therefore, the time series of the water surface elevation during the individual ship passage was identified and de-tided to obtain the water surface elevation due to ship movement alone. The frequency difference between the high-frequency ship waves and low-frequency tidal forcing made it possible to separate the ship waves from the tidal forcing. Fig. 7 confirmed the presence of high energy during the passage of the ship Guang Ping while strong absence of energy during the calm period. Therefore, we could say that the present methodology adopted in the

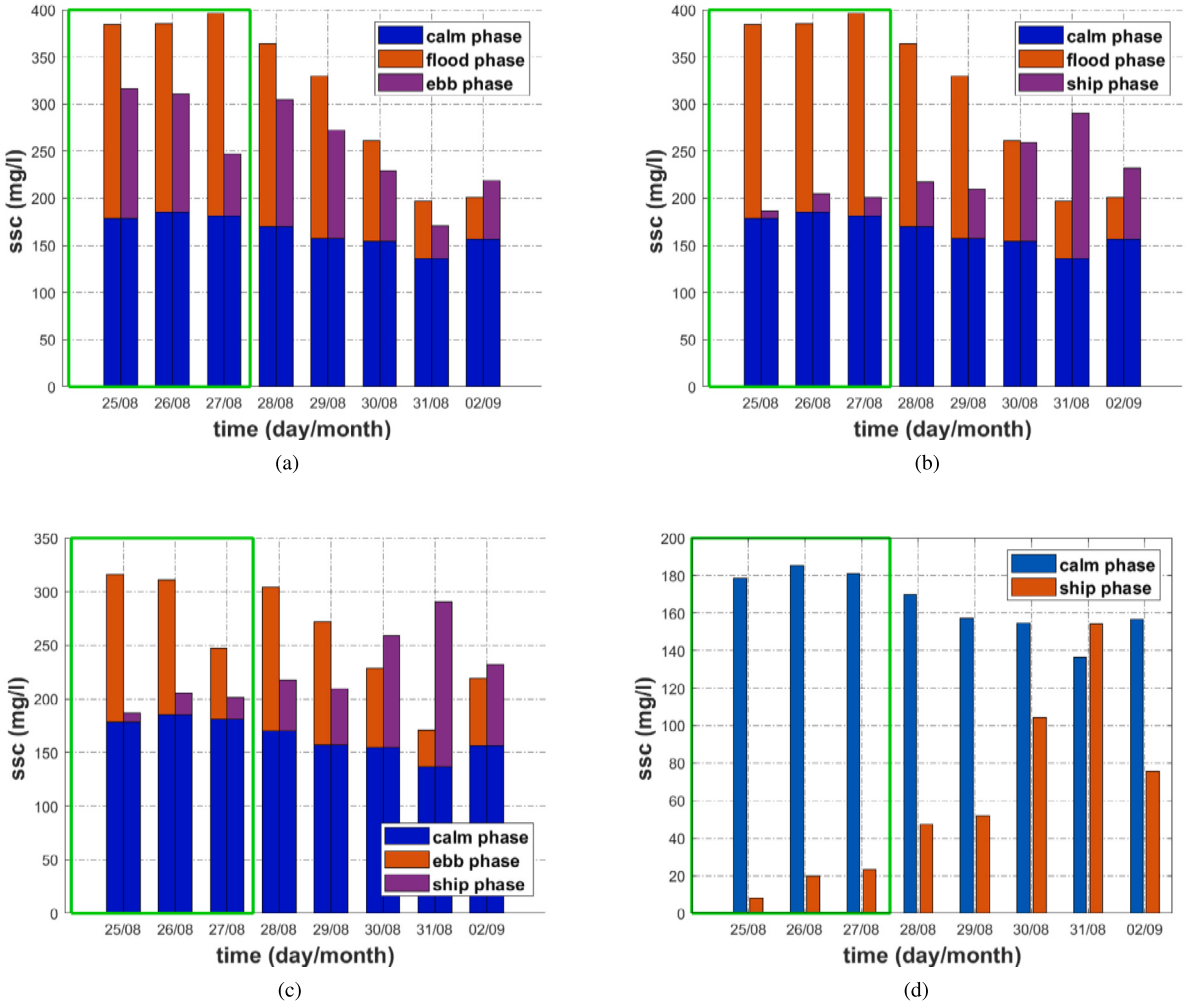


Fig. 12. Daywise comparative plot of the maximum SSC during (a) flood and ebb phase. (b) flood and ship navigation phase (c) ebb and ship navigation phase (d) calm and ship navigation phase. The green box denotes the spring period.

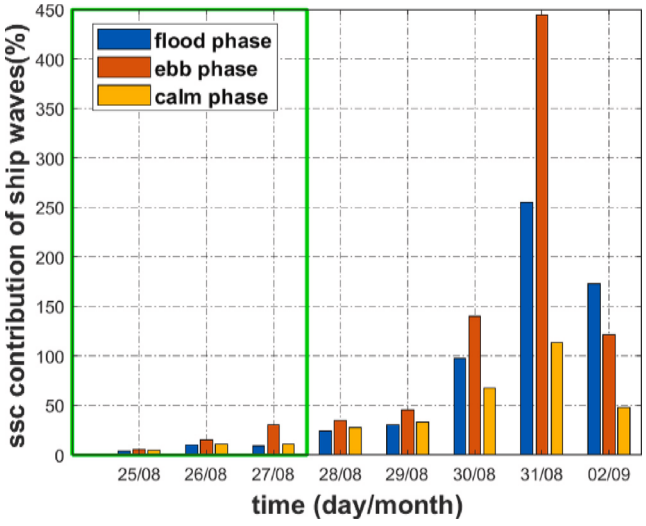


Fig. 13. Daywise percentage contribution of the max SSC due to ship waves with respect to the max SSC due to flood currents, ebb currents and upstream discharge.

present research was successful in filtering the ship-generated waves from the tides. The investigation of the filtered water surface elevation generated by the ships is reported in detail in Chakraborty et al. (2022).

It is well-known from the literature that ship-generated waves create complex hydrodynamics in shallow waterways. The ship waves feel the river bed due to limited water depth and shoaling of the ship waves induces heavy turbulence near the river bank. As a result, the bed shear stress exceeds the critical bed shear stress causing significant damage to the river banks. This adverse situation is worsened in intertidal waterways due to the combined effect of ship waves and tides. Therefore, it is necessary to quantify the sediment resuspension due to individual ship passage and with the various phases of the tide. Safak et al. (2021) provided an overall analysis of the ship-induced sediment resuspension in an intertidal waterway. It was reported in their study that the overall contribution of the tides and ship-generated waves was more during the spring tide than the neap tide. They provided the estimation of total sediment flux during boat activity and inactivity periods. Finally, they concluded that ship-generated waves contributed 12% of the sediment resuspension during their tenure of field measurements. The main research gap in their study is that individual sediment flux generated by each wake event could not be analyzed. Therefore, the investigation of the sediment flux as a function of the cardinal ship and river parameters could not be addressed. Therefore, in order to address these research gaps, systematic measurement of sediment resuspension both in spatial and temporal scales is important. Therefore, the present research adopted a new methodology to measure ship-generated waves from acoustic measurements.

The measurement of SSC can be done in various ways, such as by using conventional (laboratory analysis), optical, and acoustic methods.

Table 3
Measurement technique of SSC.

Technology	Operating principle	Advantages	Disadvantage
Water sampling	Water sediment sample is taken and analyzed in a laboratory	Reliable, informative	Flow-intrusive, Labour-intensive, Low-frequency, Poor-spatial resolution,near-bed data missing
Optical	Light backscatter through the water-sediment sample is measured and translated to SSC with calibration	High accuracy, Good spatial resolution, High frequency	Flow-intrusive, Particle-size dependent, Limited-measurement range, Uncertainties in high SSC
Acoustic	Echo strength from the sample determines SSC based on calibration	Non-intrusive, Good spatial resolution, High frequency, synchronous SSC and velocity	Low-accuracy, Limited-measurement range, Uncertainties in high SSC

The summary of the methods is listed in Table 3. The conventional method requires numerous water samples at each point and each depth of the observation station. The SSC given by this method contains a relative error of 20% from sampling and later analysis (McHenry et al., 1968). Moreover, this method has various limitations: it is unable to provide time-series data and spatial characteristics of SSC because it requires continuous water sampling as it is labour-intensive (Tessier et al., 2008; Moore et al., 2012).

Optical backscattering sensors (OBS) measure SSC by means of backscattered light, and their measurements are restricted to a single point in a fixed deployment (Lin et al., 2020). The readings of OBS are highly sensitive toward particle size distribution, and the reading of the same sensor may increase by as much as ten times for the same SSC with a smaller particle size (Ludwig and Hanes, 1990). The OBS has an upper limit because of the signal attenuation (Haught et al., 2017). In order to address these research gaps, a new methodology was adopted in the present study. The backscattering intensity information obtained from the acoustic sensors of HADCP is used extensively to calculate SSC in the Hooghly river. In addition to SSC, acoustic sensors also measure flow velocity synchronously. The HADCP could provide sufficient information both on temporal and spatial scales. However, the conversion of the backscattering intensity to SSC is not straightforward and contains significant uncertainties. Therefore, the I_{dB} obtained from the acoustic intensity was calibrated using field-measured SSC data, as reported in Appendix. The calibration equation obtained from the linear regression analysis was used to estimate the sediment resuspension due to tides and ship-generated waves.

The ships mainly navigated through the study domain during the high tide. The main assumption in the estimation of the SSC due to individual wave events was that the overall river condition is assumed to be constant during the time of an individual ship passage. In addition, sufficient time was available to reach the ambient river condition before the arrival of the next ship. Based on these assumptions, the time series of the back-scattering intensity from the HADCP during the exact time of each ship passage was separated and applied to the calibration equation to generate the SSC. Now, the main challenge was to separate the contribution of the ship waves from the total SSC obtained from the calibration equation. Since it is already known that the ship-induced SSC increases instantaneously in a matter of minutes during the ship passage and decays quickly as ship waves attenuate faster. Therefore, the application of a moving-average filter to the total SSC helped to separate the ship-induced SSC from the raw data. The time series of the SSC, as reported in Fig. 9 showed the maximum SSC generated by the ship Guang Ping is 130.96 mg/l. Now, in order to estimate the long-term effect of the ship-induced SSC on the river bank erosion and the overall contribution of the sediment flux due to ship-generated waves is important. Therefore, the present study deployed Eq. (3) to estimate SSC_flux due to each ship passage. It was found that the sediment flux generated by the ships in the spring period was significantly less than the sediment flux during the neap period. This is mainly because of the huge sediment influx in the river as complex turbulence was generated due to higher tidal fluctuations in the spring period. It was observed that the efficiency of the acoustic sensors was greatly reduced at huge sediment concentrations. In order to estimate the long-term contribution of the ship-generated waves the average sediment flux obtained during the field measurements was multiplied by the total ship traffic recorded in the calendar year 2019–2020 to get an approximate yearly estimate of the sediment flux that is carried downstream. However, it should be noted that seasonal variability plays an important role in the yearly estimate of sediment flux. Therefore, the present estimation of the total sediment flux due to ship waves may be underestimated as our measurements were performed in the monsoon period when the background SSC levels are quite significant.

As it is reported in Chakraborty et al. (2022), the non-linearity and energy contained in the ship waves increase with the increasing ship speed and irregular bathymetry, therefore, a correlation study of the SSC with the depth Froude number (which is the non-dimensional parameter obtained from ship speed and water depth) was performed. The correlation study revealed that Fr_d is highly correlated with the sediment flux with a correlation coefficient of 0.73. Fig. 10 revealed that sediment flux increases with the increasing depth Froude number. The present study performed a regression analysis on the Fr_d and SSC_{flux} to develop a new empirical equation, which could predict the sediment flux generated by the ship-generated waves. The proposed model could be widely used by the science community to get a preliminary estimation of the sediment flux generated due to ship wakes. However, it should be noted that the proposed model slightly over-predicted the sediment flux as complex ship-generated hydrodynamics could not be fully explained by using one parameter. A detailed study of the cardinal ship parameters and river conditions is required to propose an efficient model for the SSC prediction due to ship-generated waves. This was the primary limitation of the present study which is addressed in Chakraborty et al. (2023b).

5. Conclusion

The present research was focused on studying the impact of sediment transport due to ship-generated waves in intertidal waterways employing field investigation. The Falta stretch in the Hooghly river was best suited for the study due to complex hydrodynamics generated by the movement of marine vehicles, tidal currents and discharge from upstream. The ship waves travel as a series of waves, comprising both primary and secondary waves. The ship waves tend to resuspend sediments from the bed and banks due to shallow water effects. Therefore, the study of the ship waves in the intertidal environment was executed and analyzed. The separation of the water level fluctuation due to ship waves from the tidal fluctuations is challenging to assess. The time–frequency analysis of the measured water surface elevation effectively identified the ship waves from the tidal forcing due to their transient nature. The spectrogram plot reveals that ship waves appear as a chirp signal and can be easily identified from the background noise, wind waves and tidal forcing. Though the measurement of SSC using HADCP resolved the issues reported in the literature, the conversion of echo intensity to SSC has an ambiguity problem. The ADCP is highly sensitive to high concentrations of sediments and efficiency is slightly reduced. The ADCP might have slightly underestimated the velocity

data and echo intensity levels generated due to ship waves as its sampling frequency is only 2 Hz. Nevertheless, after calibration and implementation of the new methodology, the backscattering intensity of the ADCP could analyze the SSC with reasonable accuracy. The next challenge was to filter the individual effect of each ship on the sediment transport from the measured data. Contrary to tidal waves, flow velocities under ship waves can increase by order of magnitude within seconds, resulting in a less gradual, almost instantaneous increase in suspended sediment concentrations (SSC). Moreover, the SSC only increases for relatively short periods (per wave event), in order of minutes after a passing vessel, as waves attenuate much faster. Therefore, applying a moving average filter to the extracted SSC data during this short period revealed the individual contribution of ship waves.

Although the proposed methodology is justified by the presented results, the present study emphasizes that the entire characteristics of the ship-generated waves and the non-linearity associated with them could not be addressed by single-point measurements. The complex hydrodynamics generated by the ship-generated waves due to irregular bathymetry and river geometry could not be addressed by the proposed simplistic model. However, the model could be used for preliminary investigation of the induced SSC due to ship-generated waves. The natural geometry of the river not only governs the flow parameter but also governs the wave propagation along the banks. Another limitation of the present study is that the upstream discharge and upstream sediment influx were not considered in the present study. Therefore, the total sediment flux may be underestimated for the present case study. It should be noted that the present study assumed that the overall river conditions and bed morphology are constant during the passage of the ship through the study domain. However, collecting and analyzing observations on bed state change with each passing ship with wake effects on the bottom boundary layer could be a potential scope for future studies. The complex scenarios arising in the waterway due to multiple ship movements with varying dimensions and carrying capacity should be analyzed for a detailed understanding of the sediment resuspension due to ship-generated waves. The analysis of the simplistic model could be helpful for process understanding. Further research is required to understand the characteristics of ship-generated waves on sediment resuspension. The correlation study and regression analysis with important ship and river parameters could result in a robust empirical model for the prediction of ship-generated SSC. The authors have addressed these research gaps in Chakraborty et al. (2023b). However, based on the present study salient outcomes are listed below.

- The time–frequency analysis is an effective tool for separating the ship-generated waves from the tides.
- The present study emphasizes the use of acoustic intensities to analyze the sediment resuspension generated by ship waves after calibrating the instrument with in-situ site conditions.
- The comparative analysis of the SSC due to ship waves and tidal forcing during the spring and neap phases revealed that the contribution of the ship waves in the intertidal environment is noteworthy and should be considered. The tidal forcing is the major contributor to sediment resuspension during the spring phase due to high river currents and greater amplitude of tidal fluctuations.
- The SSC due to flood and ebb currents were 111.62% and 36.6% greater in the spring period than the neap period, respectively. The ship waves contribution of SSC during the spring phase is a bare minimum and is only 7.71% and 17.34% compared to the contribution of flood and ebb currents, respectively.
- The ship waves are the significant contributors of SSC during the neap phase and contribute 115.97% and 157.42% of the flood and ebb currents, respectively. The ship waves contributed 24.66% of the total SSC in the channel. In a calendar year, the approximate amount of sediments flux by ship waves is $1.46 * 10^6$ g/m^2.

It is worth mentioning that the contribution of sediment transport due to ship waves and tidal forcing is affected by ship traffic and seasonal variability. The present research was conducted during the monsoon period when the upstream discharge is highest in the waterway, suppressing the ship waves' effect to some extent. Therefore, the ship waves' contribution to sediment transport is slightly underestimated in the present study. The present study for a limited period in the Hooghly river pointed out a substantial contribution of SSC due to ship waves. Therefore, the long-term implication of the ship waves will negatively impact the stability of the waterway due to its complex hydrodynamics, resulting in bank erosion, increased turbidity and degrading the water quality. The influence of ship-generated waves and induced sediments in the bed morphology and bank erosion is a necessary investigation and will be carried out in future studies.

CRediT authorship contribution statement

Mainak Chakraborty: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Field measurements, Writing – original draft. **V. Sriram:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition, Resources. **K. Murali:** Writing – review & editing, Supervision, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request

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All authors read and approved the final manuscript.

Appendix. Conversion of acoustic intensity to SSC

ADCP instruments transmit acoustic signals to the water column and hit underwater objects like suspended sediments (Simmonds and Maclennan, 2007). The acoustic signals from the sediments were backscattered to the ADCP and are termed backscattering intensity (I_{dB}). The SSC was estimated from this backscattering intensity by means of the sonar principle. The detailed description is reported in Chakraborty et al. (2023b). However, the main steps involved in the calculation of SSC from acoustic echo intensity are presented here. Several parameters such as transmission losses due to distance, sound absorption, and acoustic wave attenuation need to be analyzed for the

conversion of acoustic data to SSC. I_{dB} is estimated by the following equation:

$$I_{dB} = k_s(I_{counts} - I_r) + 2TL \tag{5}$$

where TL is the transmission loss and I_{counts} is the backscattering intensity in counts and I_r is the noise level which is 40 counts for the present case. I_{dB} is obtained in decibels by multiplying the scaling factor k_s, given by:

$$k_s = \frac{127.3}{T + 273} \tag{6}$$

where T is the water temperature in °C. The detailed calculation of the transmission losses is reported in Chakraborty et al. (2023b).

Strong variation in suspended sediment concentration was observed in the study domain throughout the day with the variation in tides. The ADCP could collect data up to 150 m during heavy sediment influx during the flood and ebb tide. One of the limitations of the ADCP was that the acoustic signals attenuated within 150 m at high sediment concentration. Therefore, it is mandatory to calibrate the acoustic intensity with field-measured SSC. Water samples were collected at the same depth where the HADCP was mounted in the Falta jetty. The sampling was done at distances of 5 m, 10 m, 20 m, 50 m, 100 m, 150 m and 250 m from the ADCP at the same transect of ADCP installation to align the samples with the ADCP beam backscattering intensity.

$$SSC = \frac{(B - C) \times 1000}{D} \tag{7}$$

The collected water samples from the Niskin sampler were stored in 1-litre polyethylene bottles and brought to the laboratory for further analysis. Now, a filter paper was used to filter 200 ml of water, denoted by D in Eq. (7). Before filtration, the paper was appropriately dried and weighed (C). After the filtration process, the paper is again dried at 110 °C and cooled for 1 h. After the cooling process, the filter paper was weighed again (B). Now, the SSC estimation was executed by using Eq. (7) (Gartner, 2004; Dwinovantyo et al., 2017). Finally, the calibration of the acoustic intensity was executed by using linear regression analysis with the field-measured data. The equation is given by:

$$SSC = 10^{(P + Q \cdot I_{dB})} \tag{8}$$

where P and Q are the intercept and slope obtained from the linear regression analysis. The range of I_{dB} was between 20 to 80 dB and the range of SSC was between 50 to 190 mg/l.

The correlation study between the field-measured SSC and I_{dB} revealed that they are highly correlated with each other with a correlation coefficient of 0.99. The study reveals that they are positively correlated with each other. The coefficient of determination was 0.99. The linear regression analysis revealed that the coefficients P and Q from Eq. (8) are 0.0105 and 1.4638 respectively. The derived calibration equation has been applied to the present research to understand the impact of the ship waves on the sediment resuspension for the current study location.

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